Exercise 1 – QFT1

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Lattice Propagator

Consider a four-dimensional lattice with spacing Δ.

$$x = \Delta n, \qquad \text{lattice point.}$$

From every lattice point there are 8 possible links:

$$\pm e_1, \quad \pm e_2, \quad \pm e_3, \quad \pm e_4.$$

The propagator is

$$G(x) = \sum_{0 \to x} e^{-m_0 L(\text{path})},$$

where

$$L(\text{path}) = \text{number of links in the path.}$$

Directions are discrete: 8 possible directions.

Define $N_L(x)$ as the number of lattice paths from 0 to x with exactly L links. Then

$$G(x) = \sum_{L=0}^{\infty} e^{-m_0 L} N_L(x).$$

The endpoint after L links is

$$x = \Delta \sum_{k=1}^{L} \nu_k,$$

where

$$\nu_k \in \{+e_1, -e_1, +e_2, -e_2, +e_3, -e_3, +e_4, -e_4\},$$

and therefore

$$n = \sum_{k=1}^{L} \nu_k.$$

We need to count the sequences

$$\{\nu_1, \ldots, \nu_L\}$$

whose sum is n.
Thus,

$$N_L(n) = \sum_{\{\nu_1,\ldots,\nu_L\}} \delta^{(4)} \left(\sum_{k=1}^{L} \nu_k - n \right).$$

In 4-dimensions,

$$\delta^{(4)}_{n,0} = \int_{[-\pi,\pi]^4} \frac{d^4k}{(2\pi)^4} e^{ik\cdot n}.$$

Therefore,

$$N_L(n) = \int_{[-\pi,\pi]^4} \frac{d^4k}{(2\pi)^4} e^{-ik\cdot n} \sum_{\{\nu_1,\ldots,\nu_L\}} e^{ik\cdot(\nu_1+\cdots+\nu_L)}.$$

The steps are independent, so

$$\sum_{\{\nu_1,\ldots,\nu_L\}} e^{ik\cdot(\nu_1+\cdots+\nu_L)} = \left[\sum_{\nu} e^{ik\cdot\nu} \right]^L.$$

Hence,

$$N_L(n) = \int_{[-\pi,\pi]^4} \frac{d^4k}{(2\pi)^4} e^{-ik\cdot n} \left[\sum_{\nu} e^{ik\cdot\nu} \right]^L.$$

Similar to Chapter 6, let us call

$$I(k) = \sum_{\nu} e^{ik\cdot\nu},$$

where each

$$\nu = \pm e_\mu, \qquad \mu = 1, 2, 3, 4.$$

Then

$$I(k) = \sum_{\mu=1}^{4} \left(e^{ik_\mu} + e^{-ik_\mu} \right) = 2 \sum_{\mu=1}^{4} \cos k_\mu.$$

Thus, the propagator is

$$G(n) = \sum_{L=0}^{\infty} e^{-m_0 L} \int_{[-\pi,\pi]^4} \frac{d^4k}{(2\pi)^4} e^{-ik\cdot n} \left[2 \sum_{\mu=1}^{4} \cos k_\mu \right]^L.$$

Equivalently,

$$G(n) = \int_{[-\pi,\pi]^4} \frac{d^4k}{(2\pi)^4} e^{-ik\cdot n} \sum_{L=0}^{\infty} \left[2e^{-m_0} \sum_{\mu=1}^{4} \cos k_\mu \right]^L.$$

For $m_0 > \log 8$, the geometric series converges since

$$\left| 2e^{-m_0} \sum_{\mu=1}^{4} \cos k_\mu \right| \leq 8e^{-m_0} < 1.$$

Using the relation

$$\sum_{L=0}^{\infty} r^L = \frac{1}{1-r}, \qquad |r| < 1,$$

the propagator becomes

$$\boxed{G(n) = \int_{[-\pi,\pi]^4} \frac{d^4k}{(2\pi)^4} e^{-ik\cdot n} \frac{1}{1 - 2e^{-m_0} \sum_{\mu=1}^{4} \cos k_\mu}.}$$

The Continuum Limit

The continuum limit is

$$\Delta \to 0.$$

Let

$$\begin{cases} P : \text{physical momentum}, \\ k : \text{dimensionless lattice momentum}, \end{cases} \qquad k_\mu = \Delta P_\mu.$$

We have

$$I(P\Delta) = 2 \sum_{\mu=1}^{4} \cos(P_\mu \Delta),$$

and

$$\cos(P_\mu \Delta) = 1 - \frac{1}{2} P_\mu^2 \Delta^2 + O(\Delta^4).$$

Using the cosine expansion and the fact that

$$2 \sum_{\mu=1}^{4} 1 = 8,$$

we have

$$I(P\Delta) = 8 - P^2\Delta^2 + O(\Delta^4),$$

where

$$P^2 = \sum_{\mu=1}^{4} P_\mu^2.$$

Or we can write

$$I(P\Delta) = 8\left[1 - \frac{1}{8}P^2\Delta^2 + O(\Delta^4)\right].$$

Now for L links,

$$I(P\Delta)^L = 8^L\left[1 - \frac{1}{8}P^2\Delta^2 + O(\Delta^4)\right]^L.$$

Therefore,

$$\boxed{I(P\Delta)^L \sim 8^L e^{-\frac{1}{8}L\Delta^2P^2}.}$$

The relevant continuum combination is

$$L\Delta^2$$

and not $L\Delta$.
Indeed, the factor

$$e^{-\frac{1}{8}L\Delta^2P^2}$$

has a nontrivial finite limit only when

$$\boxed{L\Delta^2 = \text{finite.}}$$

Each L-link path contributes with the weight

$$e^{-m_0L}.$$

Therefore,

$$e^{-m_0L}I(P\Delta)^L \sim e^{-m_0L}8^Le^{-\frac{1}{8}L\Delta^2P^2}.$$

The first two factors can be combined as

$$8^Le^{-m_0L} = e^{-L(m_0-\log 8)}.$$

Thus,

$$e^{-m_0L}I(P\Delta)^L \sim \exp\left[-L(m_0 - \log 8) - \frac{1}{8}L\Delta^2P^2\right].$$

The relevant random-walk scaling is

$$L\Delta^2 = \text{finite.}$$

The momentum-dependent factor is

$$e^{-\frac{1}{8}L\Delta^2P^2}.$$

Define

$$s \coloneqq \frac{1}{8}L\Delta^2 \qquad \Longrightarrow \qquad L = \frac{8s}{\Delta^2}.$$

Thus,

$$L \to \infty \qquad \text{as} \qquad \Delta \to 0.$$

But these paths must contribute a finite amount:

$$e^{-L(m_0 - \log 8)}$$

must have a finite limit.
Since

$$L \sim \frac{1}{\Delta^2},$$

we require

$$m_0 - \log 8 = O(\Delta^2).$$

We want the continuum momentum-space propagator to have the denominator

$$P^2 + m^2.$$

So we choose

$$m_0(\Delta) - \log 8 = \frac{1}{8} m^2 \Delta^2 + O(\Delta^4).$$

Equivalently,

$$\boxed{m_0(\Delta) = \log 8 + \frac{1}{8} m^2 \Delta^2 + O(\Delta^4).}$$

With this choice,

$$-L(m_0 - \log 8) \longrightarrow -m^2 s,$$

and

$$-\frac{1}{8} L \Delta^2 P^2 \longrightarrow -sP^2.$$

Therefore,

$$\boxed{e^{-m_0 L} I (P\Delta)^L \longrightarrow e^{-s(m^2 + P^2)}.}$$

This is the desired factor.

Critical Value

The denominator of the propagator at $k = 0$ is

$$1 - 2e^{-m_0} \sum_{\mu=1}^{4} \cos(0).$$

Since

$$\cos(0) = 1,$$

we obtain

$$1 - 8e^{-m_0}.$$

The denominator is zero when

$$1 - 8e^{-m_0} = 0,$$

which gives

$$m_0 = \log 8.$$

Thus the critical value is

$$\boxed{m_{0,c} = \log 8.}$$

This critical value can also be understood directly from the competition between the number of lattice paths and the weight of each path:

$$8^L e^{-m_0 L} = e^{-L(m_0 - \log 8)}.$$

Therefore,

$$\begin{cases} m_0 > \log 8, & \text{long paths are exponentially suppressed,} \\ m_0 = \log 8, & \text{the path entropy and mass suppression balance,} \\ m_0 < \log 8, & \text{the number of paths overwhelms the suppression.} \end{cases}$$

Thus, in the continuum limit, the bare mass must approach the critical value according to

$$m_0(\Delta) = \log 8 + \frac{1}{8} m^2 \Delta^2 + O(\Delta^4).$$

0.1 AI use declaration

1. Model used: ChatGPT (free)

2. Revisitions/corrections made to the model output, and clarifications: I gave the model the problem and the approach needed to solve it. It gave answers mostly based on the reference notes. At some points, I have emphasized that it stays focused on the problem at hand and combine different approaches.

3. Comparison/checks against the lecture notes: It compared the solution of this problem to that of Chapter 6 (path integral for a relativistic particle).

4. Calculation steps, limiting cases, or consistency checks verified by hand: All of the calculations here are verified by hand. I did them myself, and in the end, I asked it to convert my handwritten file to LaTeX code.

5. Calculation steps, limiting cases, or consistency checks subjected to automated verification (via Mathematica, Python, LLMs): None.